DQN vs DDPG - Stat Tests

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1 Imports

[15]:
```
import numpy as np
import pandas as pd
from scipy.stats import permutation_test
import itertools
```

2 Data Loading

[16]:
```
models = [
    "logit", # row 0
    "hotelling", # row 1
    "linear" # row 2
]

firm1 = "dqn"
firm2 = "ddpg"

shock_none_data = {
    model: {
        "avg_prices": {
            firm1: [],
            firm2: []
        },
        "deltas": {
            firm1: [],
            firm2: []
        },
        "rpdis": {
            firm1: [],
            firm2: []
        }
    } for model in models
}

shock_a_data = {
```

```
    model: {
        "avg_prices": {
            firm1: [],
            firm2: []
        },
        "deltas": {
            firm1: [],
            firm2: []
        },
        "rpdis": {
            firm1: [],
            firm2: []
        }
    } for model in models
}

shock_b_data = {
    model: {
        "avg_prices": {
            firm1: [],
            firm2: []
        },
        "deltas": {
            firm1: [],
            firm2: []
        },
        "rpdis": {
            firm1: [],
            firm2: []
        }
    } for model in models
}

shock_c_data = {
    model: {
        "avg_prices": {
            firm1: [],
            firm2: []
        },
        "deltas": {
            firm1: [],
            firm2: []
        },
        "rpdis": {
            firm1: [],
            firm2: []
        }
```

```
    } for model in models
}

for i in range(1,8):
    shock_none_df = pd.read_csv(f"../results/run_{i}/shock_none/
 ↪{firm1}_vs_{firm2}.csv")
    shock_a_df = pd.read_csv(f"../results/run_{i}/shock_a/
 ↪{firm1}_vs_{firm2}_schemeA.csv")
    shock_b_df = pd.read_csv(f"../results/run_{i}/shock_b/
 ↪{firm1}_vs_{firm2}_schemeB.csv")
    shock_c_df = pd.read_csv(f"../results/run_{i}/shock_c/
 ↪{firm1}_vs_{firm2}_schemeC.csv")

    # Map each dataframe to its corresponding data storage dictionary
    shock_mappings = [
        (shock_none_df, shock_none_data),
        (shock_a_df, shock_a_data),
        (shock_b_df, shock_b_data),
        (shock_c_df, shock_c_data)
    ]

    for df, data_dict in shock_mappings:
        for idx, model in enumerate(models):
            # Access the row matching the current model by its index (0, 1, or␣
 ↪2)
            row = df.iloc[idx]

            # Dynamically look up columns based on firm name capitalization (e.
 ↪g., "Q" and "PSO")
            f1_upper = firm1.upper()
            f2_upper = firm2.upper()

            # Append Average Prices
            data_dict[model]["avg_prices"][firm1].append(row[f"{f1_upper} Avg.␣
 ↪Prices"])
            data_dict[model]["avg_prices"][firm2].append(row[f"{f2_upper} Avg.␣
 ↪Prices"])

            # Append Extra-profits Δ (deltas)
            data_dict[model]["deltas"][firm1].append(row[f"{f1_upper}␣
 ↪Extra-profits Δ"])
            data_dict[model]["deltas"][firm2].append(row[f"{f2_upper}␣
 ↪Extra-profits Δ"])

            # Append RPDI
            data_dict[model]["rpdis"][firm1].append(row[f"{f1_upper} RPDI"])
```

```
            data_dict[model]["rpdis"][firm2].append(row[f"{f2_upper} RPDI"])
```

[17]:
```
def paired_mean_diff(x, y, axis=0):
    return np.mean(x - y, axis=axis)
```

3 Logit

3.1 Shock None

3.1.1 Average Prices

[18]:
```
array1 = np.array(shock_none_data['logit']['avg_prices'][firm1])
array2 = np.array(shock_none_data['logit']['avg_prices'][firm2])

main_test = permutation_test(
    data=(array1, array2),
    statistic=paired_mean_diff,
    permutation_type='samples',     # 'samples' handles paired data structural␣
 ↪dependencies
    alternative='two-sided',
    n_resamples=np.inf
)


# --- 4. Compute the Test-Inverted 95% Confidence Interval ---
# We run a fine grid search to map exactly where the p-value transitions past 0.
 ↪05
shifts = np.linspace(-0.5, 0.5, 10001)
ci_values = []

for shift in shifts:
    # Shift sample x by a hypothetical effect size delta
    shifted_array1 = array1 - shift

    perm_res = permutation_test(
        data=(shifted_array1, array2),
        statistic=paired_mean_diff,
        permutation_type='samples',
        alternative='two-sided',
        n_resamples=np.inf
    )

    # If p > 0.05, this shift magnitude is statistically plausible (inside the␣
 ↪CI)
    if perm_res.pvalue > 0.05:
        ci_values.append(shift)
```

```python
# Extract boundaries from the valid grid values
if len(ci_values) > 0:
    lower_bound = np.min(ci_values)
    upper_bound = np.max(ci_values)
    ci_str = f"[{lower_bound:.4f}, {upper_bound:.4f}]"
else:
    ci_str = "Outside defined search grid"


# --- 5. Report Results ---
print("==========================================")
print("  EXACT PAIRED PERMUTATION TEST REPORT   ")
print("==========================================")
print(f"Observed Mean Difference: {main_test.statistic:.4f}")
print(f"Exact P-value:            {main_test.pvalue:.5f}")
print(f"95% Test-Inverted CI:     {ci_str}")
print("==========================================")
```

```
==========================================
  EXACT PAIRED PERMUTATION TEST REPORT
==========================================
Observed Mean Difference: -0.1171
Exact P-value:            0.01562
95% Test-Inverted CI:     [-0.1400, -0.1000]
==========================================
```

3.1.2 Deltas

[19]:
```python
array1 = np.array(shock_none_data['logit']['deltas'][firm1])
array2 = np.array(shock_none_data['logit']['deltas'][firm2])

main_test = permutation_test(
    data=(array1, array2),
    statistic=paired_mean_diff,
    permutation_type='samples',     # 'samples' handles paired data structural␣
 ↪dependencies
    alternative='two-sided',
    n_resamples=np.inf
)


# --- 4. Compute the Test-Inverted 95% Confidence Interval ---
# We run a fine grid search to map exactly where the p-value transitions past 0.
 ↪05
shifts = np.linspace(-1, 1, 10001)
ci_values = []
```

```python
for shift in shifts:
    # Shift sample x by a hypothetical effect size delta
    shifted_array1 = array1 - shift

    perm_res = permutation_test(
        data=(shifted_array1, array2),
        statistic=paired_mean_diff,
        permutation_type='samples',
        alternative='two-sided',
        n_resamples=np.inf
    )

    # If p > 0.05, this shift magnitude is statistically plausible (inside the CI)
    if perm_res.pvalue > 0.05:
        ci_values.append(shift)

# Extract boundaries from the valid grid values
if len(ci_values) > 0:
    lower_bound = np.min(ci_values)
    upper_bound = np.max(ci_values)
    ci_str = f"[{lower_bound:.4f}, {upper_bound:.4f}]"
else:
    ci_str = "Outside defined search grid"
    print(f"{array1 = }, {array2 = }")


# --- 5. Report Results ---
print("==========================================")
print("  EXACT PAIRED PERMUTATION TEST REPORT   ")
print("==========================================")
print(f"Observed Mean Difference: {main_test.statistic:.4f}")
print(f"Exact P-value:            {main_test.pvalue:.5f}")
print(f"95% Test-Inverted CI:     {ci_str}")
print("==========================================")
```

```
==========================================
  EXACT PAIRED PERMUTATION TEST REPORT
==========================================
Observed Mean Difference: 0.7786
Exact P-value:            0.01562
95% Test-Inverted CI:     [0.6800, 0.9200]
==========================================
```

3.1.3 RPDIs

[20]:
```
array1 = np.array(shock_none_data['logit']['rpdis'][firm1])
array2 = np.array(shock_none_data['logit']['rpdis'][firm2])

main_test = permutation_test(
    data=(array1, array2),
    statistic=paired_mean_diff,
    permutation_type='samples',      # 'samples' handles paired data structural␣
 ↪dependencies
    alternative='two-sided',
    n_resamples=np.inf
)



# --- 4. Compute the Test-Inverted 95% Confidence Interval ---
# We run a fine grid search to map exactly where the p-value transitions past 0.
 ↪05
shifts = np.linspace(-0.5, 0.5, 10001)
ci_values = []

for shift in shifts:
    # Shift sample x by a hypothetical effect size delta
    shifted_array1 = array1 - shift

    perm_res = permutation_test(
        data=(shifted_array1, array2),
        statistic=paired_mean_diff,
        permutation_type='samples',
        alternative='two-sided',
        n_resamples=np.inf
    )

    # If p > 0.05, this shift magnitude is statistically plausible (inside the␣
 ↪CI)
    if perm_res.pvalue > 0.05:
        ci_values.append(shift)

# Extract boundaries from the valid grid values
if len(ci_values) > 0:
    lower_bound = np.min(ci_values)
    upper_bound = np.max(ci_values)
    ci_str = f"[{lower_bound:.4f}, {upper_bound:.4f}]"
else:
    ci_str = "Outside defined search grid"
```

```python
# --- 5. Report Results ---
print("==========================================")
print("  EXACT PAIRED PERMUTATION TEST REPORT   ")
print("==========================================")
print(f"Observed Mean Difference: {main_test.statistic:.4f}")
print(f"Exact P-value:            {main_test.pvalue:.5f}")
print(f"95% Test-Inverted CI:     {ci_str}")
print("==========================================")
```

```
==========================================
  EXACT PAIRED PERMUTATION TEST REPORT
==========================================
Observed Mean Difference: -0.2529
Exact P-value:            0.01562
95% Test-Inverted CI:     [-0.3200, -0.2100]
==========================================
```

3.2 Shock A

3.2.1 Average Prices

[21]:
```python
array1 = np.array(shock_a_data['logit']['avg_prices'][firm1])
array2 = np.array(shock_a_data['logit']['avg_prices'][firm2])

main_test = permutation_test(
    data=(array1, array2),
    statistic=paired_mean_diff,
    permutation_type='samples',     # 'samples' handles paired data structural␣
 ↪dependencies
    alternative='two-sided',
    n_resamples=np.inf
)


# --- 4. Compute the Test-Inverted 95% Confidence Interval ---
# We run a fine grid search to map exactly where the p-value transitions past 0.
 ↪05
shifts = np.linspace(-0.5, 0.5, 10001)
ci_values = []

for shift in shifts:
    # Shift sample x by a hypothetical effect size delta
    shifted_array1 = array1 - shift

    perm_res = permutation_test(
        data=(shifted_array1, array2),
        statistic=paired_mean_diff,
        permutation_type='samples',
```

```
        alternative='two-sided',
        n_resamples=np.inf
    )

    # If p > 0.05, this shift magnitude is statistically plausible (inside the␣
↪CI)
    if perm_res.pvalue > 0.05:
        ci_values.append(shift)

# Extract boundaries from the valid grid values
if len(ci_values) > 0:
    lower_bound = np.min(ci_values)
    upper_bound = np.max(ci_values)
    ci_str = f"[{lower_bound:.4f}, {upper_bound:.4f}]"
else:
    ci_str = "Outside defined search grid"


# --- 5. Report Results ---
print("==========================================")
print("  EXACT PAIRED PERMUTATION TEST REPORT   ")
print("==========================================")
print(f"Observed Mean Difference: {main_test.statistic:.4f}")
print(f"Exact P-value:            {main_test.pvalue:.5f}")
print(f"95% Test-Inverted CI:     {ci_str}")
print("==========================================")
```

```
==========================================
  EXACT PAIRED PERMUTATION TEST REPORT
==========================================
Observed Mean Difference: -0.0871
Exact P-value:            0.01562
95% Test-Inverted CI:     [-0.1100, -0.0650]
==========================================
```

3.2.2 Deltas

[22]:
```
array1 = np.array(shock_a_data['logit']['deltas'][firm1])
array2 = np.array(shock_a_data['logit']['deltas'][firm2])

main_test = permutation_test(
    data=(array1, array2),
    statistic=paired_mean_diff,
    permutation_type='samples',     # 'samples' handles paired data structural␣
↪dependencies
    alternative='two-sided',
    n_resamples=np.inf
```

```python
)


# --- 4. Compute the Test-Inverted 95% Confidence Interval ---
# We run a fine grid search to map exactly where the p-value transitions past 0.
↪05
shifts = np.linspace(-3, 3, 10001)
ci_values = []

for shift in shifts:
    # Shift sample x by a hypothetical effect size delta
    shifted_array1 = array1 - shift

    perm_res = permutation_test(
        data=(shifted_array1, array2),
        statistic=paired_mean_diff,
        permutation_type='samples',
        alternative='two-sided',
        n_resamples=np.inf
    )

    # If p > 0.05, this shift magnitude is statistically plausible (inside the␣
↪CI)
    if perm_res.pvalue > 0.05:
        ci_values.append(shift)

# Extract boundaries from the valid grid values
if len(ci_values) > 0:
    lower_bound = np.min(ci_values)
    upper_bound = np.max(ci_values)
    ci_str = f"[{lower_bound:.4f}, {upper_bound:.4f}]"
else:
    ci_str = "Outside defined search grid"


# --- 5. Report Results ---
print("=========================================")
print("  EXACT PAIRED PERMUTATION TEST REPORT   ")
print("=========================================")
print(f"Observed Mean Difference: {main_test.statistic:.4f}")
print(f"Exact P-value:            {main_test.pvalue:.5f}")
print(f"95% Test-Inverted CI:     {ci_str}")
print("=========================================")
```

```
=========================================
  EXACT PAIRED PERMUTATION TEST REPORT
=========================================
```

```
Observed Mean Difference: 1.9114
Exact P-value:            0.01562
95% Test-Inverted CI:     [1.5252, 2.2848]
=========================================
```

3.2.3 RPDIs

[23]:

```python
array1 = np.array(shock_a_data['logit']['rpdis'][firm1])
array2 = np.array(shock_a_data['logit']['rpdis'][firm2])

main_test = permutation_test(
    data=(array1, array2),
    statistic=paired_mean_diff,
    permutation_type='samples',     # 'samples' handles paired data structural␣
 ↪dependencies
    alternative='two-sided',
    n_resamples=np.inf
)


# --- 4. Compute the Test-Inverted 95% Confidence Interval ---
# We run a fine grid search to map exactly where the p-value transitions past 0.
 ↪05
shifts = np.linspace(-0.5, 0.5, 10001)
ci_values = []

for shift in shifts:
    # Shift sample x by a hypothetical effect size delta
    shifted_array1 = array1 - shift

    perm_res = permutation_test(
        data=(shifted_array1, array2),
        statistic=paired_mean_diff,
        permutation_type='samples',
        alternative='two-sided',
        n_resamples=np.inf
    )

    # If p > 0.05, this shift magnitude is statistically plausible (inside the␣
 ↪CI)
    if perm_res.pvalue > 0.05:
        ci_values.append(shift)

# Extract boundaries from the valid grid values
if len(ci_values) > 0:
    lower_bound = np.min(ci_values)
    upper_bound = np.max(ci_values)
```

```python
    ci_str = f"[{lower_bound:.4f}, {upper_bound:.4f}]"
else:
    ci_str = "Outside defined search grid"


# --- 5. Report Results ---
print("==========================================")
print("  EXACT PAIRED PERMUTATION TEST REPORT   ")
print("==========================================")
print(f"Observed Mean Difference: {main_test.statistic:.4f}")
print(f"Exact P-value:            {main_test.pvalue:.5f}")
print(f"95% Test-Inverted CI:     {ci_str}")
print("==========================================")
```

```
==========================================
  EXACT PAIRED PERMUTATION TEST REPORT
==========================================
Observed Mean Difference: -0.3143
Exact P-value:            0.01562
95% Test-Inverted CI:     [-0.4000, -0.2250]
==========================================
```

3.3 Shock B

3.3.1 Average Prices

[24]:
```python
array1 = np.array(shock_b_data['logit']['avg_prices'][firm1])
array2 = np.array(shock_b_data['logit']['avg_prices'][firm2])

main_test = permutation_test(
    data=(array1, array2),
    statistic=paired_mean_diff,
    permutation_type='samples',     # 'samples' handles paired data structural␣
 ↪dependencies
    alternative='two-sided',
    n_resamples=np.inf
)


# --- 4. Compute the Test-Inverted 95% Confidence Interval ---
# We run a fine grid search to map exactly where the p-value transitions past 0.
 ↪05
shifts = np.linspace(-0.5, 0.5, 10001)
ci_values = []

for shift in shifts:
    # Shift sample x by a hypothetical effect size delta
    shifted_array1 = array1 - shift
```

```python
    perm_res = permutation_test(
        data=(shifted_array1, array2),
        statistic=paired_mean_diff,
        permutation_type='samples',
        alternative='two-sided',
        n_resamples=np.inf
    )

    # If p > 0.05, this shift magnitude is statistically plausible (inside the␣
↪CI)
    if perm_res.pvalue > 0.05:
        ci_values.append(shift)

# Extract boundaries from the valid grid values
if len(ci_values) > 0:
    lower_bound = np.min(ci_values)
    upper_bound = np.max(ci_values)
    ci_str = f"[{lower_bound:.4f}, {upper_bound:.4f}]"
else:
    ci_str = "Outside defined search grid"


# --- 5. Report Results ---
print("==========================================")
print("  EXACT PAIRED PERMUTATION TEST REPORT   ")
print("==========================================")
print(f"Observed Mean Difference: {main_test.statistic:.4f}")
print(f"Exact P-value:            {main_test.pvalue:.5f}")
print(f"95% Test-Inverted CI:     {ci_str}")
print("==========================================")
```

```
==========================================
  EXACT PAIRED PERMUTATION TEST REPORT
==========================================
Observed Mean Difference: -0.0943
Exact P-value:            0.01562
95% Test-Inverted CI:     [-0.1200, -0.0700]
==========================================
```

3.3.2 Deltas

[25]:
```python
array1 = np.array(shock_b_data['logit']['deltas'][firm1])
array2 = np.array(shock_b_data['logit']['deltas'][firm2])

main_test = permutation_test(
    data=(array1, array2),
    statistic=paired_mean_diff,
```

```python
    permutation_type='samples',     # 'samples' handles paired data structural␣
↪dependencies
    alternative='two-sided',
    n_resamples=np.inf
)


# --- 4. Compute the Test-Inverted 95% Confidence Interval ---
# We run a fine grid search to map exactly where the p-value transitions past 0.
↪05
shifts = np.linspace(-1, 1, 10001)
ci_values = []

for shift in shifts:
    # Shift sample x by a hypothetical effect size delta
    shifted_array1 = array1 - shift

    perm_res = permutation_test(
        data=(shifted_array1, array2),
        statistic=paired_mean_diff,
        permutation_type='samples',
        alternative='two-sided',
        n_resamples=np.inf
    )

    # If p > 0.05, this shift magnitude is statistically plausible (inside the␣
↪CI)
    if perm_res.pvalue > 0.05:
        ci_values.append(shift)

# Extract boundaries from the valid grid values
if len(ci_values) > 0:
    lower_bound = np.min(ci_values)
    upper_bound = np.max(ci_values)
    ci_str = f"[{lower_bound:.4f}, {upper_bound:.4f}]"
else:
    ci_str = "Outside defined search grid"


# --- 5. Report Results ---
print("==========================================")
print("  EXACT PAIRED PERMUTATION TEST REPORT   ")
print("==========================================")
print(f"Observed Mean Difference: {main_test.statistic:.4f}")
print(f"Exact P-value:            {main_test.pvalue:.5f}")
print(f"95% Test-Inverted CI:     {ci_str}")
print("==========================================")
```

```
==========================================
  EXACT PAIRED PERMUTATION TEST REPORT
==========================================
Observed Mean Difference: 0.8343
Exact P-value:            0.01562
95% Test-Inverted CI:     [0.6300, 1.0000]
==========================================
```

3.3.3 RPDIs

[26]:
```
array1 = np.array(shock_b_data['logit']['rpdis'][firm1])
array2 = np.array(shock_b_data['logit']['rpdis'][firm2])

main_test = permutation_test(
    data=(array1, array2),
    statistic=paired_mean_diff,
    permutation_type='samples',     # 'samples' handles paired data structural␣
 ↪dependencies
    alternative='two-sided',
    n_resamples=np.inf
)


# --- 4. Compute the Test-Inverted 95% Confidence Interval ---
# We run a fine grid search to map exactly where the p-value transitions past 0.
 ↪05
shifts = np.linspace(-0.5, 0.5, 10001)
ci_values = []

for shift in shifts:
    # Shift sample x by a hypothetical effect size delta
    shifted_array1 = array1 - shift

    perm_res = permutation_test(
        data=(shifted_array1, array2),
        statistic=paired_mean_diff,
        permutation_type='samples',
        alternative='two-sided',
        n_resamples=np.inf
    )

    # If p > 0.05, this shift magnitude is statistically plausible (inside the␣
 ↪CI)
    if perm_res.pvalue > 0.05:
        ci_values.append(shift)

# Extract boundaries from the valid grid values
```

```
if len(ci_values) > 0:
    lower_bound = np.min(ci_values)
    upper_bound = np.max(ci_values)
    ci_str = f"[{lower_bound:.4f}, {upper_bound:.4f}]"
else:
    ci_str = "Outside defined search grid"


# --- 5. Report Results ---
print("=========================================")
print("  EXACT PAIRED PERMUTATION TEST REPORT   ")
print("=========================================")
print(f"Observed Mean Difference: {main_test.statistic:.4f}")
print(f"Exact P-value:            {main_test.pvalue:.5f}")
print(f"95% Test-Inverted CI:     {ci_str}")
print("=========================================")
```

```
=========================================
  EXACT PAIRED PERMUTATION TEST REPORT
=========================================
Observed Mean Difference: -0.2343
Exact P-value:            0.01562
95% Test-Inverted CI:     [-0.3050, -0.1650]
=========================================
```

3.4 Shock C

3.4.1 Average Prices

[27]:
```
array1 = np.array(shock_c_data['logit']['avg_prices'][firm1])
array2 = np.array(shock_c_data['logit']['avg_prices'][firm2])

main_test = permutation_test(
    data=(array1, array2),
    statistic=paired_mean_diff,
    permutation_type='samples',      # 'samples' handles paired data structural␣
 ↪dependencies
    alternative='two-sided',
    n_resamples=np.inf
)


# --- 4. Compute the Test-Inverted 95% Confidence Interval ---
# We run a fine grid search to map exactly where the p-value transitions past 0.
 ↪05
shifts = np.linspace(-0.5, 0.5, 10001)
ci_values = []
```

```python
for shift in shifts:
    # Shift sample x by a hypothetical effect size delta
    shifted_array1 = array1 - shift

    perm_res = permutation_test(
        data=(shifted_array1, array2),
        statistic=paired_mean_diff,
        permutation_type='samples',
        alternative='two-sided',
        n_resamples=np.inf
    )

    # If p > 0.05, this shift magnitude is statistically plausible (inside the CI)
    if perm_res.pvalue > 0.05:
        ci_values.append(shift)

# Extract boundaries from the valid grid values
if len(ci_values) > 0:
    lower_bound = np.min(ci_values)
    upper_bound = np.max(ci_values)
    ci_str = f"[{lower_bound:.4f}, {upper_bound:.4f}]"
else:
    ci_str = "Outside defined search grid"


# --- 5. Report Results ---
print("=========================================")
print("  EXACT PAIRED PERMUTATION TEST REPORT   ")
print("=========================================")
print(f"Observed Mean Difference: {main_test.statistic:.4f}")
print(f"Exact P-value:            {main_test.pvalue:.5f}")
print(f"95% Test-Inverted CI:     {ci_str}")
print("=========================================")
```

```
=========================================
  EXACT PAIRED PERMUTATION TEST REPORT
=========================================
Observed Mean Difference: -0.0214
Exact P-value:            0.18750
95% Test-Inverted CI:     [-0.0550, 0.0050]
=========================================
```

3.4.2 Deltas

[28]:
```
array1 = np.array(shock_c_data['logit']['deltas'][firm1])
array2 = np.array(shock_c_data['logit']['deltas'][firm2])

main_test = permutation_test(
    data=(array1, array2),
    statistic=paired_mean_diff,
    permutation_type='samples',     # 'samples' handles paired data structural␣
↪dependencies
    alternative='two-sided',
    n_resamples=np.inf
)



# --- 4. Compute the Test-Inverted 95% Confidence Interval ---
# We run a fine grid search to map exactly where the p-value transitions past 0.
↪05
shifts = np.linspace(-1.5, 1.5, 10001)
ci_values = []

for shift in shifts:
    # Shift sample x by a hypothetical effect size delta
    shifted_array1 = array1 - shift

    perm_res = permutation_test(
        data=(shifted_array1, array2),
        statistic=paired_mean_diff,
        permutation_type='samples',
        alternative='two-sided',
        n_resamples=np.inf
    )

    # If p > 0.05, this shift magnitude is statistically plausible (inside the␣
↪CI)
    if perm_res.pvalue > 0.05:
        ci_values.append(shift)

# Extract boundaries from the valid grid values
if len(ci_values) > 0:
    lower_bound = np.min(ci_values)
    upper_bound = np.max(ci_values)
    ci_str = f"[{lower_bound:.4f}, {upper_bound:.4f}]"
else:
    ci_str = "Outside defined search grid"
```

```python
# --- 5. Report Results ---
print("==========================================")
print("  EXACT PAIRED PERMUTATION TEST REPORT   ")
print("==========================================")
print(f"Observed Mean Difference: {main_test.statistic:.4f}")
print(f"Exact P-value:            {main_test.pvalue:.5f}")
print(f"95% Test-Inverted CI:     {ci_str}")
print("==========================================")
```

```
==========================================
  EXACT PAIRED PERMUTATION TEST REPORT
==========================================
Observed Mean Difference: 1.2557
Exact P-value:            0.01562
95% Test-Inverted CI:     [0.7302, 1.5000]
==========================================
```

3.4.3 RPDIs

[29]:
```python
array1 = np.array(shock_c_data['logit']['rpdis'][firm1])
array2 = np.array(shock_c_data['logit']['rpdis'][firm2])

main_test = permutation_test(
    data=(array1, array2),
    statistic=paired_mean_diff,
    permutation_type='samples',     # 'samples' handles paired data structural␣
 ↪dependencies
    alternative='two-sided',
    n_resamples=np.inf
)


# --- 4. Compute the Test-Inverted 95% Confidence Interval ---
# We run a fine grid search to map exactly where the p-value transitions past 0.
 ↪05
shifts = np.linspace(-0.5, 0.5, 10001)
ci_values = []

for shift in shifts:
    # Shift sample x by a hypothetical effect size delta
    shifted_array1 = array1 - shift

    perm_res = permutation_test(
        data=(shifted_array1, array2),
        statistic=paired_mean_diff,
        permutation_type='samples',
        alternative='two-sided',
```

```
        n_resamples=np.inf
    )

    # If p > 0.05, this shift magnitude is statistically plausible (inside the␣
↪CI)
    if perm_res.pvalue > 0.05:
        ci_values.append(shift)

# Extract boundaries from the valid grid values
if len(ci_values) > 0:
    lower_bound = np.min(ci_values)
    upper_bound = np.max(ci_values)
    ci_str = f"[{lower_bound:.4f}, {upper_bound:.4f}]"
else:
    ci_str = "Outside defined search grid"


# --- 5. Report Results ---
print("==========================================")
print("  EXACT PAIRED PERMUTATION TEST REPORT   ")
print("==========================================")
print(f"Observed Mean Difference: {main_test.statistic:.4f}")
print(f"Exact P-value:            {main_test.pvalue:.5f}")
print(f"95% Test-Inverted CI:     {ci_str}")
print("==========================================")
```

```
==========================================
  EXACT PAIRED PERMUTATION TEST REPORT
==========================================
Observed Mean Difference: -0.0771
Exact P-value:            0.17188
95% Test-Inverted CI:     [-0.2050, 0.0150]
==========================================
```

4 Hotelling

4.1 Shock None

4.1.1 Average Prices

[30]:
```
array1 = np.array(shock_none_data['hotelling']['avg_prices'][firm1])
array2 = np.array(shock_none_data['hotelling']['avg_prices'][firm2])

main_test = permutation_test(
    data=(array1, array2),
    statistic=paired_mean_diff,
    permutation_type='samples',     # 'samples' handles paired data structural␣
↪dependencies
```

```python
    alternative='two-sided',
    n_resamples=np.inf
)


# --- 4. Compute the Test-Inverted 95% Confidence Interval ---
# We run a fine grid search to map exactly where the p-value transitions past 0.
↪05
shifts = np.linspace(-0.5, 0.5, 10001)
ci_values = []

for shift in shifts:
    # Shift sample x by a hypothetical effect size delta
    shifted_array1 = array1 - shift

    perm_res = permutation_test(
        data=(shifted_array1, array2),
        statistic=paired_mean_diff,
        permutation_type='samples',
        alternative='two-sided',
        n_resamples=np.inf
    )

    # If p > 0.05, this shift magnitude is statistically plausible (inside the␣
↪CI)
    if perm_res.pvalue > 0.05:
        ci_values.append(shift)

# Extract boundaries from the valid grid values
if len(ci_values) > 0:
    lower_bound = np.min(ci_values)
    upper_bound = np.max(ci_values)
    ci_str = f"[{lower_bound:.4f}, {upper_bound:.4f}]"
else:
    ci_str = "Outside defined search grid"
    print(f"{array1 = }, {array2 = }")


# --- 5. Report Results ---
print("=========================================")
print("  EXACT PAIRED PERMUTATION TEST REPORT   ")
print("=========================================")
print(f"Observed Mean Difference: {main_test.statistic:.4f}")
print(f"Exact P-value:            {main_test.pvalue:.5f}")
print(f"95% Test-Inverted CI:     {ci_str}")
print("=========================================")
```

```
==========================================
  EXACT PAIRED PERMUTATION TEST REPORT
==========================================
Observed Mean Difference: -0.0143
Exact P-value:            0.18750
95% Test-Inverted CI:     [-0.0400, 0.0100]
==========================================
```

4.1.2 Deltas

[31]:
```python
array1 = np.array(shock_none_data['hotelling']['deltas'][firm1])
array2 = np.array(shock_none_data['hotelling']['deltas'][firm2])

main_test = permutation_test(
    data=(array1, array2),
    statistic=paired_mean_diff,
    permutation_type='samples',     # 'samples' handles paired data structural␣
 ↪dependencies
    alternative='two-sided',
    n_resamples=np.inf
)


# --- 4. Compute the Test-Inverted 95% Confidence Interval ---
# We run a fine grid search to map exactly where the p-value transitions past 0.
 ↪05
shifts = np.linspace(-0.5, 0.5, 10001)
ci_values = []

for shift in shifts:
    # Shift sample x by a hypothetical effect size delta
    shifted_array1 = array1 - shift

    perm_res = permutation_test(
        data=(shifted_array1, array2),
        statistic=paired_mean_diff,
        permutation_type='samples',
        alternative='two-sided',
        n_resamples=np.inf
    )

    # If p > 0.05, this shift magnitude is statistically plausible (inside the␣
 ↪CI)
    if perm_res.pvalue > 0.05:
        ci_values.append(shift)

# Extract boundaries from the valid grid values
```

```python
if len(ci_values) > 0:
    lower_bound = np.min(ci_values)
    upper_bound = np.max(ci_values)
    ci_str = f"[{lower_bound:.4f}, {upper_bound:.4f}]"
else:
    ci_str = "Outside defined search grid"


# --- 5. Report Results ---
print("=========================================")
print("  EXACT PAIRED PERMUTATION TEST REPORT   ")
print("=========================================")
print(f"Observed Mean Difference: {main_test.statistic:.4f}")
print(f"Exact P-value:            {main_test.pvalue:.5f}")
print(f"95% Test-Inverted CI:     {ci_str}")
print("=========================================")
```

```
=========================================
  EXACT PAIRED PERMUTATION TEST REPORT
=========================================
Observed Mean Difference: 0.0886
Exact P-value:            0.10938
95% Test-Inverted CI:     [-0.0250, 0.2000]
=========================================
```

4.1.3 RPDIs

[32]:
```python
array1 = np.array(shock_none_data['hotelling']['rpdis'][firm1])
array2 = np.array(shock_none_data['hotelling']['rpdis'][firm2])

main_test = permutation_test(
    data=(array1, array2),
    statistic=paired_mean_diff,
    permutation_type='samples',     # 'samples' handles paired data structural␣
 ↪dependencies
    alternative='two-sided',
    n_resamples=np.inf
)


# --- 4. Compute the Test-Inverted 95% Confidence Interval ---
# We run a fine grid search to map exactly where the p-value transitions past 0.
 ↪05
shifts = np.linspace(-0.5, 0.5, 10001)
ci_values = []

for shift in shifts:
```

```python
    # Shift sample x by a hypothetical effect size delta
    shifted_array1 = array1 - shift

    perm_res = permutation_test(
        data=(shifted_array1, array2),
        statistic=paired_mean_diff,
        permutation_type='samples',
        alternative='two-sided',
        n_resamples=np.inf
    )

    # If p > 0.05, this shift magnitude is statistically plausible (inside the␣
↪CI)
    if perm_res.pvalue > 0.05:
        ci_values.append(shift)

# Extract boundaries from the valid grid values
if len(ci_values) > 0:
    lower_bound = np.min(ci_values)
    upper_bound = np.max(ci_values)
    ci_str = f"[{lower_bound:.4f}, {upper_bound:.4f}]"
else:
    ci_str = "Outside defined search grid"


# --- 5. Report Results ---
print("==========================================")
print("  EXACT PAIRED PERMUTATION TEST REPORT   ")
print("==========================================")
print(f"Observed Mean Difference: {main_test.statistic:.4f}")
print(f"Exact P-value:            {main_test.pvalue:.5f}")
print(f"95% Test-Inverted CI:     {ci_str}")
print("==========================================")
```

```
==========================================
  EXACT PAIRED PERMUTATION TEST REPORT
==========================================
Observed Mean Difference: -0.0343
Exact P-value:            0.39062
95% Test-Inverted CI:     [-0.1200, 0.0600]
==========================================
```

4.2 Shock A

4.2.1 Average Prices

[33]:
```
array1 = np.array(shock_a_data['hotelling']['avg_prices'][firm1])
array2 = np.array(shock_a_data['hotelling']['avg_prices'][firm2])

main_test = permutation_test(
    data=(array1, array2),
    statistic=paired_mean_diff,
    permutation_type='samples',     # 'samples' handles paired data structural␣
 ↪dependencies
    alternative='two-sided',
    n_resamples=np.inf
)


# --- 4. Compute the Test-Inverted 95% Confidence Interval ---
# We run a fine grid search to map exactly where the p-value transitions past 0.
 ↪05
shifts = np.linspace(-0.5, 0.5, 10001)
ci_values = []

for shift in shifts:
    # Shift sample x by a hypothetical effect size delta
    shifted_array1 = array1 - shift

    perm_res = permutation_test(
        data=(shifted_array1, array2),
        statistic=paired_mean_diff,
        permutation_type='samples',
        alternative='two-sided',
        n_resamples=np.inf
    )

    # If p > 0.05, this shift magnitude is statistically plausible (inside the␣
 ↪CI)
    if perm_res.pvalue > 0.05:
        ci_values.append(shift)

# Extract boundaries from the valid grid values
if len(ci_values) > 0:
    lower_bound = np.min(ci_values)
    upper_bound = np.max(ci_values)
    ci_str = f"[{lower_bound:.4f}, {upper_bound:.4f}]"
else:
    ci_str = "Outside defined search grid"
```

```
# --- 5. Report Results ---
print("==========================================")
print("  EXACT PAIRED PERMUTATION TEST REPORT   ")
print("==========================================")
print(f"Observed Mean Difference: {main_test.statistic:.4f}")
print(f"Exact P-value:            {main_test.pvalue:.5f}")
print(f"95% Test-Inverted CI:     {ci_str}")
print("==========================================")
```

```
==========================================
  EXACT PAIRED PERMUTATION TEST REPORT
==========================================
Observed Mean Difference: -0.0043
Exact P-value:            0.62500
95% Test-Inverted CI:     [-0.0250, 0.0150]
==========================================
```

4.2.2 Deltas

[34]:
```
array1 = np.array(shock_a_data['hotelling']['deltas'][firm1])
array2 = np.array(shock_a_data['hotelling']['deltas'][firm2])

main_test = permutation_test(
    data=(array1, array2),
    statistic=paired_mean_diff,
    permutation_type='samples',     # 'samples' handles paired data structural␣
 ↪dependencies
    alternative='two-sided',
    n_resamples=np.inf
)


# --- 4. Compute the Test-Inverted 95% Confidence Interval ---
# We run a fine grid search to map exactly where the p-value transitions past 0.
 ↪05
shifts = np.linspace(-0.5, 0.5, 10001)
ci_values = []

for shift in shifts:
    # Shift sample x by a hypothetical effect size delta
    shifted_array1 = array1 - shift

    perm_res = permutation_test(
        data=(shifted_array1, array2),
        statistic=paired_mean_diff,
        permutation_type='samples',
```

```
        alternative='two-sided',
        n_resamples=np.inf
    )

    # If p > 0.05, this shift magnitude is statistically plausible (inside the␣
↪CI)
    if perm_res.pvalue > 0.05:
        ci_values.append(shift)

# Extract boundaries from the valid grid values
if len(ci_values) > 0:
    lower_bound = np.min(ci_values)
    upper_bound = np.max(ci_values)
    ci_str = f"[{lower_bound:.4f}, {upper_bound:.4f}]"
else:
    ci_str = "Outside defined search grid"


# --- 5. Report Results ---
print("=========================================")
print("  EXACT PAIRED PERMUTATION TEST REPORT   ")
print("=========================================")
print(f"Observed Mean Difference: {main_test.statistic:.4f}")
print(f"Exact P-value:            {main_test.pvalue:.5f}")
print(f"95% Test-Inverted CI:     {ci_str}")
print("=========================================")
```

```
=========================================
  EXACT PAIRED PERMUTATION TEST REPORT
=========================================
Observed Mean Difference: 0.0643
Exact P-value:            0.03125
95% Test-Inverted CI:     [0.0100, 0.1300]
=========================================
```

4.2.3 RPDIs

[35]:
```
array1 = np.array(shock_a_data['hotelling']['rpdis'][firm1])
array2 = np.array(shock_a_data['hotelling']['rpdis'][firm2])

main_test = permutation_test(
    data=(array1, array2),
    statistic=paired_mean_diff,
    permutation_type='samples',     # 'samples' handles paired data structural␣
↪dependencies
    alternative='two-sided',
    n_resamples=np.inf
```

```python
)


# --- 4. Compute the Test-Inverted 95% Confidence Interval ---
# We run a fine grid search to map exactly where the p-value transitions past 0.
↪05
shifts = np.linspace(-0.5, 0.5, 10001)
ci_values = []

for shift in shifts:
    # Shift sample x by a hypothetical effect size delta
    shifted_array1 = array1 - shift

    perm_res = permutation_test(
        data=(shifted_array1, array2),
        statistic=paired_mean_diff,
        permutation_type='samples',
        alternative='two-sided',
        n_resamples=np.inf
    )

    # If p > 0.05, this shift magnitude is statistically plausible (inside the
↪CI)
    if perm_res.pvalue > 0.05:
        ci_values.append(shift)

# Extract boundaries from the valid grid values
if len(ci_values) > 0:
    lower_bound = np.min(ci_values)
    upper_bound = np.max(ci_values)
    ci_str = f"[{lower_bound:.4f}, {upper_bound:.4f}]"
else:
    ci_str = "Outside defined search grid"


# --- 5. Report Results ---
print("==========================================")
print("  EXACT PAIRED PERMUTATION TEST REPORT   ")
print("==========================================")
print(f"Observed Mean Difference: {main_test.statistic:.4f}")
print(f"Exact P-value:            {main_test.pvalue:.5f}")
print(f"95% Test-Inverted CI:     {ci_str}")
print("==========================================")
```

```
==========================================
  EXACT PAIRED PERMUTATION TEST REPORT
==========================================
```

```
Observed Mean Difference: -0.0186
Exact P-value:            0.64062
95% Test-Inverted CI:     [-0.1100, 0.0600]
=========================================
```

4.3 Shock B

4.3.1 Average Prices

[36]:
```
array1 = np.array(shock_b_data['hotelling']['avg_prices'][firm1])
array2 = np.array(shock_b_data['hotelling']['avg_prices'][firm2])

main_test = permutation_test(
    data=(array1, array2),
    statistic=paired_mean_diff,
    permutation_type='samples',     # 'samples' handles paired data structural␣
 ↪dependencies
    alternative='two-sided',
    n_resamples=np.inf
)


# --- 4. Compute the Test-Inverted 95% Confidence Interval ---
# We run a fine grid search to map exactly where the p-value transitions past 0.
 ↪05
shifts = np.linspace(-0.5, 0.5, 10001)
ci_values = []

for shift in shifts:
    # Shift sample x by a hypothetical effect size delta
    shifted_array1 = array1 - shift

    perm_res = permutation_test(
        data=(shifted_array1, array2),
        statistic=paired_mean_diff,
        permutation_type='samples',
        alternative='two-sided',
        n_resamples=np.inf
    )

    # If p > 0.05, this shift magnitude is statistically plausible (inside the␣
 ↪CI)
    if perm_res.pvalue > 0.05:
        ci_values.append(shift)

# Extract boundaries from the valid grid values
if len(ci_values) > 0:
    lower_bound = np.min(ci_values)
```

```
    upper_bound = np.max(ci_values)
    ci_str = f"[{lower_bound:.4f}, {upper_bound:.4f}]"
else:
    ci_str = "Outside defined search grid"
    print(f"{array1 = }, {array2 = }")


# --- 5. Report Results ---
print("==========================================")
print("  EXACT PAIRED PERMUTATION TEST REPORT   ")
print("==========================================")
print(f"Observed Mean Difference: {main_test.statistic:.4f}")
print(f"Exact P-value:            {main_test.pvalue:.5f}")
print(f"95% Test-Inverted CI:     {ci_str}")
print("==========================================")
```

```
==========================================
  EXACT PAIRED PERMUTATION TEST REPORT
==========================================
Observed Mean Difference: 0.0186
Exact P-value:            0.06250
95% Test-Inverted CI:     [0.0000, 0.0350]
==========================================
```

4.3.2 Deltas

[37]:
```
array1 = np.array(shock_b_data['hotelling']['deltas'][firm1])
array2 = np.array(shock_b_data['hotelling']['deltas'][firm2])

main_test = permutation_test(
    data=(array1, array2),
    statistic=paired_mean_diff,
    permutation_type='samples',     # 'samples' handles paired data structural␣
 ↪dependencies
    alternative='two-sided',
    n_resamples=np.inf
)


# --- 4. Compute the Test-Inverted 95% Confidence Interval ---
# We run a fine grid search to map exactly where the p-value transitions past 0.
 ↪05
shifts = np.linspace(-0.5, 0.5, 10001)
ci_values = []

for shift in shifts:
    # Shift sample x by a hypothetical effect size delta
```

```
    shifted_array1 = array1 - shift

    perm_res = permutation_test(
        data=(shifted_array1, array2),
        statistic=paired_mean_diff,
        permutation_type='samples',
        alternative='two-sided',
        n_resamples=np.inf
    )

    # If p > 0.05, this shift magnitude is statistically plausible (inside the␣
 ↪CI)
    if perm_res.pvalue > 0.05:
        ci_values.append(shift)

# Extract boundaries from the valid grid values
if len(ci_values) > 0:
    lower_bound = np.min(ci_values)
    upper_bound = np.max(ci_values)
    ci_str = f"[{lower_bound:.4f}, {upper_bound:.4f}]"
else:
    ci_str = "Outside defined search grid"


# --- 5. Report Results ---
print("==========================================")
print("  EXACT PAIRED PERMUTATION TEST REPORT   ")
print("==========================================")
print(f"Observed Mean Difference: {main_test.statistic:.4f}")
print(f"Exact P-value:            {main_test.pvalue:.5f}")
print(f"95% Test-Inverted CI:     {ci_str}")
print("==========================================")
```

```
==========================================
  EXACT PAIRED PERMUTATION TEST REPORT
==========================================
Observed Mean Difference: -0.0071
Exact P-value:            0.81250
95% Test-Inverted CI:     [-0.0700, 0.0450]
==========================================
```

4.3.3 RPDIs

[38]:
```
array1 = np.array(shock_b_data['hotelling']['rpdis'][firm1])
array2 = np.array(shock_b_data['hotelling']['rpdis'][firm2])

main_test = permutation_test(
    data=(array1, array2),
```

```python
    statistic=paired_mean_diff,
    permutation_type='samples',     # 'samples' handles paired data structural dependencies
    alternative='two-sided',
    n_resamples=np.inf
)


# --- 4. Compute the Test-Inverted 95% Confidence Interval ---
# We run a fine grid search to map exactly where the p-value transitions past 0.05
shifts = np.linspace(-0.5, 0.5, 10001)
ci_values = []

for shift in shifts:
    # Shift sample x by a hypothetical effect size delta
    shifted_array1 = array1 - shift

    perm_res = permutation_test(
        data=(shifted_array1, array2),
        statistic=paired_mean_diff,
        permutation_type='samples',
        alternative='two-sided',
        n_resamples=np.inf
    )

    # If p > 0.05, this shift magnitude is statistically plausible (inside the CI)
    if perm_res.pvalue > 0.05:
        ci_values.append(shift)

# Extract boundaries from the valid grid values
if len(ci_values) > 0:
    lower_bound = np.min(ci_values)
    upper_bound = np.max(ci_values)
    ci_str = f"[{lower_bound:.4f}, {upper_bound:.4f}]"
else:
    ci_str = "Outside defined search grid"


# --- 5. Report Results ---
print("==========================================")
print("  EXACT PAIRED PERMUTATION TEST REPORT   ")
print("==========================================")
print(f"Observed Mean Difference: {main_test.statistic:.4f}")
print(f"Exact P-value:            {main_test.pvalue:.5f}")
print(f"95% Test-Inverted CI:     {ci_str}")
```

```
print("=========================================")
```

```
=========================================
  EXACT PAIRED PERMUTATION TEST REPORT
=========================================
Observed Mean Difference: 0.0543
Exact P-value:            0.12500
95% Test-Inverted CI:     [-0.0300, 0.1300]
=========================================
```

4.4 Shock C

4.4.1 Average Prices

[39]:
```
array1 = np.array(shock_c_data['hotelling']['avg_prices'][firm1])
array2 = np.array(shock_c_data['hotelling']['avg_prices'][firm2])

main_test = permutation_test(
    data=(array1, array2),
    statistic=paired_mean_diff,
    permutation_type='samples',      # 'samples' handles paired data structural␣
 ↪dependencies
    alternative='two-sided',
    n_resamples=np.inf
)



# --- 4. Compute the Test-Inverted 95% Confidence Interval ---
# We run a fine grid search to map exactly where the p-value transitions past 0.
 ↪05
shifts = np.linspace(-0.5, 0.5, 10001)
ci_values = []

for shift in shifts:
    # Shift sample x by a hypothetical effect size delta
    shifted_array1 = array1 - shift

    perm_res = permutation_test(
        data=(shifted_array1, array2),
        statistic=paired_mean_diff,
        permutation_type='samples',
        alternative='two-sided',
        n_resamples=np.inf
    )

    # If p > 0.05, this shift magnitude is statistically plausible (inside the␣
 ↪CI)
    if perm_res.pvalue > 0.05:
```

```
        ci_values.append(shift)

# Extract boundaries from the valid grid values
if len(ci_values) > 0:
    lower_bound = np.min(ci_values)
    upper_bound = np.max(ci_values)
    ci_str = f"[{lower_bound:.4f}, {upper_bound:.4f}]"
else:
    ci_str = "Outside defined search grid"


# --- 5. Report Results ---
print("=========================================")
print("  EXACT PAIRED PERMUTATION TEST REPORT   ")
print("=========================================")
print(f"Observed Mean Difference: {main_test.statistic:.4f}")
print(f"Exact P-value:            {main_test.pvalue:.5f}")
print(f"95% Test-Inverted CI:     {ci_str}")
print("=========================================")
```

```
=========================================
  EXACT PAIRED PERMUTATION TEST REPORT
=========================================
Observed Mean Difference: -0.0114
Exact P-value:            0.12500
95% Test-Inverted CI:     [-0.0250, 0.0000]
=========================================
```

4.4.2 Deltas

[40]:
```
array1 = np.array(shock_c_data['hotelling']['deltas'][firm1])
array2 = np.array(shock_c_data['hotelling']['deltas'][firm2])

main_test = permutation_test(
    data=(array1, array2),
    statistic=paired_mean_diff,
    permutation_type='samples',     # 'samples' handles paired data structural␣
 ↪dependencies
    alternative='two-sided',
    n_resamples=np.inf
)


# --- 4. Compute the Test-Inverted 95% Confidence Interval ---
# We run a fine grid search to map exactly where the p-value transitions past 0.
 ↪05
shifts = np.linspace(-0.5, 0.5, 10001)
```

```python
ci_values = []

for shift in shifts:
    # Shift sample x by a hypothetical effect size delta
    shifted_array1 = array1 - shift

    perm_res = permutation_test(
        data=(shifted_array1, array2),
        statistic=paired_mean_diff,
        permutation_type='samples',
        alternative='two-sided',
        n_resamples=np.inf
    )

    # If p > 0.05, this shift magnitude is statistically plausible (inside the␣
↪CI)
    if perm_res.pvalue > 0.05:
        ci_values.append(shift)

# Extract boundaries from the valid grid values
if len(ci_values) > 0:
    lower_bound = np.min(ci_values)
    upper_bound = np.max(ci_values)
    ci_str = f"[{lower_bound:.4f}, {upper_bound:.4f}]"
else:
    ci_str = "Outside defined search grid"


# --- 5. Report Results ---
print("=========================================")
print("  EXACT PAIRED PERMUTATION TEST REPORT   ")
print("=========================================")
print(f"Observed Mean Difference: {main_test.statistic:.4f}")
print(f"Exact P-value:            {main_test.pvalue:.5f}")
print(f"95% Test-Inverted CI:     {ci_str}")
print("=========================================")
```

```
=========================================
  EXACT PAIRED PERMUTATION TEST REPORT
=========================================
Observed Mean Difference: 0.0771
Exact P-value:            0.12500
95% Test-Inverted CI:     [-0.0400, 0.1850]
=========================================
```

4.4.3 RPDIs

[41]:
```
array1 = np.array(shock_c_data['hotelling']['rpdis'][firm1])
array2 = np.array(shock_c_data['hotelling']['rpdis'][firm2])

main_test = permutation_test(
    data=(array1, array2),
    statistic=paired_mean_diff,
    permutation_type='samples',     # 'samples' handles paired data structural␣
 ↪dependencies
    alternative='two-sided',
    n_resamples=np.inf
)



# --- 4. Compute the Test-Inverted 95% Confidence Interval ---
# We run a fine grid search to map exactly where the p-value transitions past 0.
 ↪05
shifts = np.linspace(-0.5, 0.5, 10001)
ci_values = []

for shift in shifts:
    # Shift sample x by a hypothetical effect size delta
    shifted_array1 = array1 - shift

    perm_res = permutation_test(
        data=(shifted_array1, array2),
        statistic=paired_mean_diff,
        permutation_type='samples',
        alternative='two-sided',
        n_resamples=np.inf
    )

    # If p > 0.05, this shift magnitude is statistically plausible (inside the␣
 ↪CI)
    if perm_res.pvalue > 0.05:
        ci_values.append(shift)

# Extract boundaries from the valid grid values
if len(ci_values) > 0:
    lower_bound = np.min(ci_values)
    upper_bound = np.max(ci_values)
    ci_str = f"[{lower_bound:.4f}, {upper_bound:.4f}]"
else:
    ci_str = "Outside defined search grid"
```

```
# --- 5. Report Results ---
print("==========================================")
print("  EXACT PAIRED PERMUTATION TEST REPORT   ")
print("==========================================")
print(f"Observed Mean Difference: {main_test.statistic:.4f}")
print(f"Exact P-value:            {main_test.pvalue:.5f}")
print(f"95% Test-Inverted CI:     {ci_str}")
print("==========================================")
```

```
==========================================
  EXACT PAIRED PERMUTATION TEST REPORT
==========================================
Observed Mean Difference: -0.0314
Exact P-value:            0.18750
95% Test-Inverted CI:     [-0.0850, 0.0150]
==========================================
```

5 Linear

5.1 Shock None

5.1.1 Average Prices

[42]:

```
array1 = np.array(shock_none_data['linear']['avg_prices'][firm1])
array2 = np.array(shock_none_data['linear']['avg_prices'][firm2])

main_test = permutation_test(
    data=(array1, array2),
    statistic=paired_mean_diff,
    permutation_type='samples',     # 'samples' handles paired data structural␣
↪dependencies
    alternative='two-sided',
    n_resamples=np.inf
)


# --- 4. Compute the Test-Inverted 95% Confidence Interval ---
# We run a fine grid search to map exactly where the p-value transitions past 0.
↪05
shifts = np.linspace(-0.5, 0.5, 10001)
ci_values = []

for shift in shifts:
    # Shift sample x by a hypothetical effect size delta
    shifted_array1 = array1 - shift

    perm_res = permutation_test(
        data=(shifted_array1, array2),
```

```
        statistic=paired_mean_diff,
        permutation_type='samples',
        alternative='two-sided',
        n_resamples=np.inf
    )

    # If p > 0.05, this shift magnitude is statistically plausible (inside the␣
↪CI)
    if perm_res.pvalue > 0.05:
        ci_values.append(shift)

# Extract boundaries from the valid grid values
if len(ci_values) > 0:
    lower_bound = np.min(ci_values)
    upper_bound = np.max(ci_values)
    ci_str = f"[{lower_bound:.4f}, {upper_bound:.4f}]"
else:
    ci_str = "Outside defined search grid"


# --- 5. Report Results ---
print("==========================================")
print("  EXACT PAIRED PERMUTATION TEST REPORT   ")
print("==========================================")
print(f"Observed Mean Difference: {main_test.statistic:.4f}")
print(f"Exact P-value:            {main_test.pvalue:.5f}")
print(f"95% Test-Inverted CI:     {ci_str}")
print("==========================================")
```

```
==========================================
  EXACT PAIRED PERMUTATION TEST REPORT
==========================================
Observed Mean Difference: 0.0029
Exact P-value:            0.50000
95% Test-Inverted CI:     [0.0000, 0.0100]
==========================================
```

5.1.2 Deltas

[43]:
```
array1 = np.array(shock_none_data['linear']['deltas'][firm1])
array2 = np.array(shock_none_data['linear']['deltas'][firm2])

main_test = permutation_test(
    data=(array1, array2),
    statistic=paired_mean_diff,
    permutation_type='samples',     # 'samples' handles paired data structural␣
↪dependencies
```

```python
    alternative='two-sided',
    n_resamples=np.inf
)

# --- 4. Compute the Test-Inverted 95% Confidence Interval ---
# We run a fine grid search to map exactly where the p-value transitions past 0.
↪05
shifts = np.linspace(-0.5, 0.5, 10001)
ci_values = []

for shift in shifts:
    # Shift sample x by a hypothetical effect size delta
    shifted_array1 = array1 - shift

    perm_res = permutation_test(
        data=(shifted_array1, array2),
        statistic=paired_mean_diff,
        permutation_type='samples',
        alternative='two-sided',
        n_resamples=np.inf
    )

    # If p > 0.05, this shift magnitude is statistically plausible (inside the␣
↪CI)
    if perm_res.pvalue > 0.05:
        ci_values.append(shift)

# Extract boundaries from the valid grid values
if len(ci_values) > 0:
    lower_bound = np.min(ci_values)
    upper_bound = np.max(ci_values)
    ci_str = f"[{lower_bound:.4f}, {upper_bound:.4f}]"
else:
    ci_str = "Outside defined search grid"
    print(f"{array1 = }, {array2 = }")


# --- 5. Report Results ---
print("==========================================")
print("  EXACT PAIRED PERMUTATION TEST REPORT    ")
print("==========================================")
print(f"Observed Mean Difference: {main_test.statistic:.4f}")
print(f"Exact P-value:            {main_test.pvalue:.5f}")
print(f"95% Test-Inverted CI:     {ci_str}")
print("==========================================")
```

```
==========================================
```

```
  EXACT PAIRED PERMUTATION TEST REPORT
=========================================
Observed Mean Difference: 0.0214
Exact P-value:            0.04688
95% Test-Inverted CI:     [0.0050, 0.0350]
=========================================
```

5.1.3 RPDIs

[44]:
```
array1 = np.array(shock_none_data['linear']['rpdis'][firm1])
array2 = np.array(shock_none_data['linear']['rpdis'][firm2])

main_test = permutation_test(
    data=(array1, array2),
    statistic=paired_mean_diff,
    permutation_type='samples',     # 'samples' handles paired data structural␣
 ↪dependencies
    alternative='two-sided',
    n_resamples=np.inf
)

# --- 4. Compute the Test-Inverted 95% Confidence Interval ---
# We run a fine grid search to map exactly where the p-value transitions past 0.
 ↪05
shifts = np.linspace(-0.5, 0.5, 10001)
ci_values = []

for shift in shifts:
    # Shift sample x by a hypothetical effect size delta
    shifted_array1 = array1 - shift

    perm_res = permutation_test(
        data=(shifted_array1, array2),
        statistic=paired_mean_diff,
        permutation_type='samples',
        alternative='two-sided',
        n_resamples=np.inf
    )

    # If p > 0.05, this shift magnitude is statistically plausible (inside the␣
 ↪CI)
    if perm_res.pvalue > 0.05:
        ci_values.append(shift)

# Extract boundaries from the valid grid values
if len(ci_values) > 0:
    lower_bound = np.min(ci_values)
```

```
    upper_bound = np.max(ci_values)
    ci_str = f"[{lower_bound:.4f}, {upper_bound:.4f}]"
else:
    ci_str = "Outside defined search grid"
    print(f"{array1 = }, {array2 = }")


# --- 5. Report Results ---
print("==========================================")
print("  EXACT PAIRED PERMUTATION TEST REPORT   ")
print("==========================================")
print(f"Observed Mean Difference: {main_test.statistic:.4f}")
print(f"Exact P-value:            {main_test.pvalue:.5f}")
print(f"95% Test-Inverted CI:     {ci_str}")
print("==========================================")
```

```
==========================================
  EXACT PAIRED PERMUTATION TEST REPORT
==========================================
Observed Mean Difference: -0.0171
Exact P-value:            0.62500
95% Test-Inverted CI:     [-0.0850, 0.0500]
==========================================
```

5.2 Shock A

5.2.1 Average Prices

[45]:
```
array1 = np.array(shock_a_data['linear']['avg_prices'][firm1])
array2 = np.array(shock_a_data['linear']['avg_prices'][firm2])

main_test = permutation_test(
    data=(array1, array2),
    statistic=paired_mean_diff,
    permutation_type='samples',     # 'samples' handles paired data structural␣
 ↪dependencies
    alternative='two-sided',
    n_resamples=np.inf
)

# --- 4. Compute the Test-Inverted 95% Confidence Interval ---
# We run a fine grid search to map exactly where the p-value transitions past 0.
 ↪05
shifts = np.linspace(-0.5, 0.5, 10001)
ci_values = []

for shift in shifts:
    # Shift sample x by a hypothetical effect size delta
```

```
    shifted_array1 = array1 - shift

    perm_res = permutation_test(
        data=(shifted_array1, array2),
        statistic=paired_mean_diff,
        permutation_type='samples',
        alternative='two-sided',
        n_resamples=np.inf
    )

    # If p > 0.05, this shift magnitude is statistically plausible (inside the␣
↪CI)
    if perm_res.pvalue > 0.05:
        ci_values.append(shift)

# Extract boundaries from the valid grid values
if len(ci_values) > 0:
    lower_bound = np.min(ci_values)
    upper_bound = np.max(ci_values)
    ci_str = f"[{lower_bound:.4f}, {upper_bound:.4f}]"
else:
    ci_str = "Outside defined search grid"
    print(f"{array1 = }, {array2 = }")


# --- 5. Report Results ---
print("==========================================")
print("  EXACT PAIRED PERMUTATION TEST REPORT   ")
print("==========================================")
print(f"Observed Mean Difference: {main_test.statistic:.4f}")
print(f"Exact P-value:            {main_test.pvalue:.5f}")
print(f"95% Test-Inverted CI:     {ci_str}")
print("==========================================")
```

```
==========================================
  EXACT PAIRED PERMUTATION TEST REPORT
==========================================
Observed Mean Difference: 0.0043
Exact P-value:            0.25000
95% Test-Inverted CI:     [0.0000, 0.0100]
==========================================
```

5.2.2 Deltas

[46]:
```
array1 = np.array(shock_a_data['linear']['deltas'][firm1])
array2 = np.array(shock_a_data['linear']['deltas'][firm2])

main_test = permutation_test(
```

```
    data=(array1, array2),
    statistic=paired_mean_diff,
    permutation_type='samples',      # 'samples' handles paired data structural␣
 ↪dependencies
    alternative='two-sided',
    n_resamples=np.inf
)

# --- 4. Compute the Test-Inverted 95% Confidence Interval ---
# We run a fine grid search to map exactly where the p-value transitions past 0.
 ↪05
shifts = np.linspace(-0.5, 0.5, 10001)
ci_values = []

for shift in shifts:
    # Shift sample x by a hypothetical effect size delta
    shifted_array1 = array1 - shift

    perm_res = permutation_test(
        data=(shifted_array1, array2),
        statistic=paired_mean_diff,
        permutation_type='samples',
        alternative='two-sided',
        n_resamples=np.inf
    )

    # If p > 0.05, this shift magnitude is statistically plausible (inside the␣
 ↪CI)
    if perm_res.pvalue > 0.05:
        ci_values.append(shift)

# Extract boundaries from the valid grid values
if len(ci_values) > 0:
    lower_bound = np.min(ci_values)
    upper_bound = np.max(ci_values)
    ci_str = f"[{lower_bound:.4f}, {upper_bound:.4f}]"
else:
    ci_str = "Outside defined search grid"
    print(f"{array1 = }, {array2 = }")


# --- 5. Report Results ---
print("==========================================")
print("  EXACT PAIRED PERMUTATION TEST REPORT    ")
print("==========================================")
print(f"Observed Mean Difference: {main_test.statistic:.4f}")
print(f"Exact P-value:            {main_test.pvalue:.5f}")
```

```
print(f"95% Test-Inverted CI:     {ci_str}")
print("=========================================")
```

```
=========================================
  EXACT PAIRED PERMUTATION TEST REPORT
=========================================
Observed Mean Difference: -0.0086
Exact P-value:            0.75000
95% Test-Inverted CI:     [-0.0700, 0.0400]
=========================================
```

5.2.3 RPDIs

[47]:
```
array1 = np.array(shock_a_data['linear']['rpdis'][firm1])
array2 = np.array(shock_a_data['linear']['rpdis'][firm2])

main_test = permutation_test(
    data=(array1, array2),
    statistic=paired_mean_diff,
    permutation_type='samples',      # 'samples' handles paired data structural␣
 ↪dependencies
    alternative='two-sided',
    n_resamples=np.inf
)

# --- 4. Compute the Test-Inverted 95% Confidence Interval ---
# We run a fine grid search to map exactly where the p-value transitions past 0.
 ↪05
shifts = np.linspace(-0.5, 0.5, 10001)
ci_values = []

for shift in shifts:
    # Shift sample x by a hypothetical effect size delta
    shifted_array1 = array1 - shift

    perm_res = permutation_test(
        data=(shifted_array1, array2),
        statistic=paired_mean_diff,
        permutation_type='samples',
        alternative='two-sided',
        n_resamples=np.inf
    )

    # If p > 0.05, this shift magnitude is statistically plausible (inside the␣
 ↪CI)
    if perm_res.pvalue > 0.05:
        ci_values.append(shift)
```

```
# Extract boundaries from the valid grid values
if len(ci_values) > 0:
    lower_bound = np.min(ci_values)
    upper_bound = np.max(ci_values)
    ci_str = f"[{lower_bound:.4f}, {upper_bound:.4f}]"
else:
    ci_str = "Outside defined search grid"
    print(f"{array1 = }, {array2 = }")


# --- 5. Report Results ---
print("==========================================")
print("  EXACT PAIRED PERMUTATION TEST REPORT   ")
print("==========================================")
print(f"Observed Mean Difference: {main_test.statistic:.4f}")
print(f"Exact P-value:            {main_test.pvalue:.5f}")
print(f"95% Test-Inverted CI:     {ci_str}")
print("==========================================")
```

```
==========================================
  EXACT PAIRED PERMUTATION TEST REPORT
==========================================
Observed Mean Difference: -0.0157
Exact P-value:            0.59375
95% Test-Inverted CI:     [-0.0900, 0.0600]
==========================================
```

5.3 Shock B

5.3.1 Average Prices

[48]:
```
array1 = np.array(shock_b_data['linear']['avg_prices'][firm1])
array2 = np.array(shock_b_data['linear']['avg_prices'][firm2])

main_test = permutation_test(
    data=(array1, array2),
    statistic=paired_mean_diff,
    permutation_type='samples',      # 'samples' handles paired data structural␣
 ↪dependencies
    alternative='two-sided',
    n_resamples=np.inf
)

# --- 4. Compute the Test-Inverted 95% Confidence Interval ---
# We run a fine grid search to map exactly where the p-value transitions past 0.
 ↪05
shifts = np.linspace(-0.5, 0.5, 10001)
```

```python
ci_values = []

for shift in shifts:
    # Shift sample x by a hypothetical effect size delta
    shifted_array1 = array1 - shift

    perm_res = permutation_test(
        data=(shifted_array1, array2),
        statistic=paired_mean_diff,
        permutation_type='samples',
        alternative='two-sided',
        n_resamples=np.inf
    )

    # If p > 0.05, this shift magnitude is statistically plausible (inside the CI)
    if perm_res.pvalue > 0.05:
        ci_values.append(shift)

# Extract boundaries from the valid grid values
if len(ci_values) > 0:
    lower_bound = np.min(ci_values)
    upper_bound = np.max(ci_values)
    ci_str = f"[{lower_bound:.4f}, {upper_bound:.4f}]"
else:
    ci_str = "Outside defined search grid"
    print(f"{array1 = }, {array2 = }")


# --- 5. Report Results ---
print("==========================================")
print("  EXACT PAIRED PERMUTATION TEST REPORT   ")
print("==========================================")
print(f"Observed Mean Difference: {main_test.statistic:.4f}")
print(f"Exact P-value:            {main_test.pvalue:.5f}")
print(f"95% Test-Inverted CI:     {ci_str}")
print("==========================================")
```

```
==========================================
  EXACT PAIRED PERMUTATION TEST REPORT
==========================================
Observed Mean Difference: 0.0057
Exact P-value:            0.12500
95% Test-Inverted CI:     [0.0000, 0.0100]
==========================================
```

5.3.2 Deltas

[49]:
```python
array1 = np.array(shock_b_data['linear']['deltas'][firm1])
array2 = np.array(shock_b_data['linear']['deltas'][firm2])

main_test = permutation_test(
    data=(array1, array2),
    statistic=paired_mean_diff,
    permutation_type='samples',     # 'samples' handles paired data structural
 ↪dependencies
    alternative='two-sided',
    n_resamples=np.inf
)

# --- 4. Compute the Test-Inverted 95% Confidence Interval ---
# We run a fine grid search to map exactly where the p-value transitions past 0.
 ↪05
shifts = np.linspace(-0.5, 0.5, 10001)
ci_values = []

for shift in shifts:
    # Shift sample x by a hypothetical effect size delta
    shifted_array1 = array1 - shift

    perm_res = permutation_test(
        data=(shifted_array1, array2),
        statistic=paired_mean_diff,
        permutation_type='samples',
        alternative='two-sided',
        n_resamples=np.inf
    )

    # If p > 0.05, this shift magnitude is statistically plausible (inside the
 ↪CI)
    if perm_res.pvalue > 0.05:
        ci_values.append(shift)

# Extract boundaries from the valid grid values
if len(ci_values) > 0:
    lower_bound = np.min(ci_values)
    upper_bound = np.max(ci_values)
    ci_str = f"[{lower_bound:.4f}, {upper_bound:.4f}]"
else:
    ci_str = "Outside defined search grid"
    print(f"{array1 = }, {array2 = }")
```

```python
# --- 5. Report Results ---
print("==========================================")
print("  EXACT PAIRED PERMUTATION TEST REPORT   ")
print("==========================================")
print(f"Observed Mean Difference: {main_test.statistic:.4f}")
print(f"Exact P-value:            {main_test.pvalue:.5f}")
print(f"95% Test-Inverted CI:     {ci_str}")
print("==========================================")
```

```
==========================================
  EXACT PAIRED PERMUTATION TEST REPORT
==========================================
Observed Mean Difference: 0.0486
Exact P-value:            0.17188
95% Test-Inverted CI:     [-0.0600, 0.1300]
==========================================
```

5.3.3 RPDIs

[50]:
```python
array1 = np.array(shock_b_data['linear']['rpdis'][firm1])
array2 = np.array(shock_b_data['linear']['rpdis'][firm2])

main_test = permutation_test(
    data=(array1, array2),
    statistic=paired_mean_diff,
    permutation_type='samples',     # 'samples' handles paired data structural␣
 ↪dependencies
    alternative='two-sided',
    n_resamples=np.inf
)

# --- 4. Compute the Test-Inverted 95% Confidence Interval ---
# We run a fine grid search to map exactly where the p-value transitions past 0.
 ↪05
shifts = np.linspace(-0.5, 0.5, 10001)
ci_values = []

for shift in shifts:
    # Shift sample x by a hypothetical effect size delta
    shifted_array1 = array1 - shift

    perm_res = permutation_test(
        data=(shifted_array1, array2),
        statistic=paired_mean_diff,
        permutation_type='samples',
        alternative='two-sided',
        n_resamples=np.inf
```

```
    )

    # If p > 0.05, this shift magnitude is statistically plausible (inside the␣
↪CI)
    if perm_res.pvalue > 0.05:
        ci_values.append(shift)

# Extract boundaries from the valid grid values
if len(ci_values) > 0:
    lower_bound = np.min(ci_values)
    upper_bound = np.max(ci_values)
    ci_str = f"[{lower_bound:.4f}, {upper_bound:.4f}]"
else:
    ci_str = "Outside defined search grid"
    print(f"{array1 = }, {array2 = }")


# --- 5. Report Results ---
print("==========================================")
print("  EXACT PAIRED PERMUTATION TEST REPORT   ")
print("==========================================")
print(f"Observed Mean Difference: {main_test.statistic:.4f}")
print(f"Exact P-value:            {main_test.pvalue:.5f}")
print(f"95% Test-Inverted CI:     {ci_str}")
print("==========================================")
```

```
==========================================
  EXACT PAIRED PERMUTATION TEST REPORT
==========================================
Observed Mean Difference: 0.0286
Exact P-value:            0.25000
95% Test-Inverted CI:     [-0.0200, 0.0900]
==========================================
```

5.4 Shock C

5.4.1 Average Prices

[51]:
```
array1 = np.array(shock_c_data['linear']['avg_prices'][firm1])
array2 = np.array(shock_c_data['linear']['avg_prices'][firm2])

main_test = permutation_test(
    data=(array1, array2),
    statistic=paired_mean_diff,
    permutation_type='samples',     # 'samples' handles paired data structural␣
↪dependencies
    alternative='two-sided',
    n_resamples=np.inf
```

```python
)

# --- 4. Compute the Test-Inverted 95% Confidence Interval ---
# We run a fine grid search to map exactly where the p-value transitions past 0.
↪05
shifts = np.linspace(-0.5, 0.5, 10001)
ci_values = []

for shift in shifts:
    # Shift sample x by a hypothetical effect size delta
    shifted_array1 = array1 - shift

    perm_res = permutation_test(
        data=(shifted_array1, array2),
        statistic=paired_mean_diff,
        permutation_type='samples',
        alternative='two-sided',
        n_resamples=np.inf
    )

    # If p > 0.05, this shift magnitude is statistically plausible (inside the␣
↪CI)
    if perm_res.pvalue > 0.05:
        ci_values.append(shift)

# Extract boundaries from the valid grid values
if len(ci_values) > 0:
    lower_bound = np.min(ci_values)
    upper_bound = np.max(ci_values)
    ci_str = f"[{lower_bound:.4f}, {upper_bound:.4f}]"
else:
    ci_str = "Outside defined search grid"
    print(f"{array1 = }, {array2 = }")


# --- 5. Report Results ---
print("=========================================")
print("  EXACT PAIRED PERMUTATION TEST REPORT   ")
print("=========================================")
print(f"Observed Mean Difference: {main_test.statistic:.4f}")
print(f"Exact P-value:            {main_test.pvalue:.5f}")
print(f"95% Test-Inverted CI:     {ci_str}")
print("=========================================")
```

```
=========================================
  EXACT PAIRED PERMUTATION TEST REPORT
=========================================
```

```
Observed Mean Difference: 0.0029
Exact P-value:            0.50000
95% Test-Inverted CI:     [0.0000, 0.0100]
=========================================
```

5.4.2 Deltas

[52]:
```python
array1 = np.array(shock_c_data['linear']['deltas'][firm1])
array2 = np.array(shock_c_data['linear']['deltas'][firm2])

main_test = permutation_test(
    data=(array1, array2),
    statistic=paired_mean_diff,
    permutation_type='samples',     # 'samples' handles paired data structural␣
 ↪dependencies
    alternative='two-sided',
    n_resamples=np.inf
)

# --- 4. Compute the Test-Inverted 95% Confidence Interval ---
# We run a fine grid search to map exactly where the p-value transitions past 0.
 ↪05
shifts = np.linspace(-0.5, 0.5, 10001)
ci_values = []

for shift in shifts:
    # Shift sample x by a hypothetical effect size delta
    shifted_array1 = array1 - shift

    perm_res = permutation_test(
        data=(shifted_array1, array2),
        statistic=paired_mean_diff,
        permutation_type='samples',
        alternative='two-sided',
        n_resamples=np.inf
    )

    # If p > 0.05, this shift magnitude is statistically plausible (inside the␣
 ↪CI)
    if perm_res.pvalue > 0.05:
        ci_values.append(shift)

# Extract boundaries from the valid grid values
if len(ci_values) > 0:
    lower_bound = np.min(ci_values)
    upper_bound = np.max(ci_values)
    ci_str = f"[{lower_bound:.4f}, {upper_bound:.4f}]"
```

```python
else:
    ci_str = "Outside defined search grid"
    print(f"{array1 = }, {array2 = }")


# --- 5. Report Results ---
print("==========================================")
print("  EXACT PAIRED PERMUTATION TEST REPORT   ")
print("==========================================")
print(f"Observed Mean Difference: {main_test.statistic:.4f}")
print(f"Exact P-value:            {main_test.pvalue:.5f}")
print(f"95% Test-Inverted CI:     {ci_str}")
print("==========================================")
```

```
==========================================
  EXACT PAIRED PERMUTATION TEST REPORT
==========================================
Observed Mean Difference: -0.0771
Exact P-value:            0.39062
95% Test-Inverted CI:     [-0.2200, 0.1050]
==========================================
```

5.4.3 RPDIs

[53]:
```python
array1 = np.array(shock_c_data['linear']['rpdis'][firm1])
array2 = np.array(shock_c_data['linear']['rpdis'][firm2])

main_test = permutation_test(
    data=(array1, array2),
    statistic=paired_mean_diff,
    permutation_type='samples',      # 'samples' handles paired data structural␣
 ↪dependencies
    alternative='two-sided',
    n_resamples=np.inf
)

# --- 4. Compute the Test-Inverted 95% Confidence Interval ---
# We run a fine grid search to map exactly where the p-value transitions past 0.
 ↪05
shifts = np.linspace(-0.5, 0.5, 10001)
ci_values = []

for shift in shifts:
    # Shift sample x by a hypothetical effect size delta
    shifted_array1 = array1 - shift

    perm_res = permutation_test(
```

```python
        data=(shifted_array1, array2),
        statistic=paired_mean_diff,
        permutation_type='samples',
        alternative='two-sided',
        n_resamples=np.inf
    )

    # If p > 0.05, this shift magnitude is statistically plausible (inside the␣
↪CI)
    if perm_res.pvalue > 0.05:
        ci_values.append(shift)

# Extract boundaries from the valid grid values
if len(ci_values) > 0:
    lower_bound = np.min(ci_values)
    upper_bound = np.max(ci_values)
    ci_str = f"[{lower_bound:.4f}, {upper_bound:.4f}]"
else:
    ci_str = "Outside defined search grid"
    print(f"{array1 = }, {array2 = }")


# --- 5. Report Results ---
print("=========================================")
print("  EXACT PAIRED PERMUTATION TEST REPORT   ")
print("=========================================")
print(f"Observed Mean Difference: {main_test.statistic:.4f}")
print(f"Exact P-value:            {main_test.pvalue:.5f}")
print(f"95% Test-Inverted CI:     {ci_str}")
print("=========================================")
```

```
=========================================
  EXACT PAIRED PERMUTATION TEST REPORT
=========================================
Observed Mean Difference: -0.0271
Exact P-value:            0.25000
95% Test-Inverted CI:     [-0.0700, 0.0200]
=========================================
```